

NADEN MOHAMED YASEN

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SUMMARY

AI & Computer Science graduate (Helwan University, 2026) with hands-on experience developing, testing, and evaluating AI tools and LLM-based systems. Proven ability to build evaluation harnesses for trackable metrics, compare modeling approaches, and communicate findings to non-technical stakeholders. Proficient with Hugging Face, LangChain, LangGraph, and various LLM APIs, driven by a commitment to testing assumptions and understanding AI capabilities.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence · Helwan University, Cairo · GPA: 3.37/4.0 Sep 2022 – Jun 2026

SKILLS

AI & LLM Evaluation: Model Benchmarking & Evaluation, LLM-as-Judge, LLMs, RAG, Agentic Systems, Prompt Engineering, Fine-Tuning (LoRA/PEFT), NLP, Deep Learning

AI Tools & Platforms: Hugging Face, LangChain, LangGraph, Streamlit, Gradio, Weights & Biases, Google Gemini, Groq API

Research & Analysis: Statistics, Model Evaluation, Exploratory Data Analysis, Data Visualization (Matplotlib, Seaborn, Plotly, Dash), Technical Reporting, MS Office (Word, PowerPoint, Excel)

Programming & Systems: Python, SQL, C, C++, Java, FastAPI, Git/GitHub, Docker, Postman, Postgres, pgvector, MongoDB Atlas, FAISS, Qdrant

Languages Spoken: Arabic, English

RELEVANT PROJECTS

ScholarFlow AI - RAG-based Document Intelligence Platform | *Python, FastAPI, PostgreSQL/pgvector, Qdrant, Docker*

- Built a RAG backend in FastAPI with a swappable provider architecture (LLM: Groq; vector DB: pgvector/Qdrant), including a PDF ingestion pipeline that preserves document structure for accurate retrieval.
- Designed the PostgreSQL schema, managed migrations, and containerized the full 7-service system with Docker Compose; independently debugged and fixed production-grade issues across the stack.
- Built an offline evaluation harness for the RAG pipeline, combining deterministic retrieval metrics (hit rate, MRR) against a golden question set to catch hallucination and grounding issues.

Adaptive Code Intelligence Assistant | *LLM Evaluation, Pydantic, pytest*

- Developed an evaluation harness to measure router classification accuracy and end-to-end code-generation correctness (pass@1) against a benchmark dataset, transforming LLM output quality into trackable metrics.
- Implemented a self-learning RAG loop with LLM-as-judge relevance checks, validated by a pytest suite testing routing accuracy, retrieval-evaluator agreement, and fallback behavior under simulated failures.
- Designed a multi-step LLM pipeline with strict Pydantic schemas at every stage, keeping component handoffs reliable and testable.

Egypt Tourism Sentiment Analysis | *TensorFlow, XGBoost, LangChain, Dash*

- Benchmarked ML approaches on 30k+ scraped tourism reviews, comparing Logistic Regression (89.6% accuracy) against XGBoost (90.7% accuracy) to identify the optimal model.
- Integrated the Google Gemini Flash API with LangChain to translate model output into actionable, business-facing recommendations.
- Communicated findings to non-technical stakeholders through confusion matrices, word clouds, and an interactive Dash dashboard.

Toxic Content Classification System | *PyTorch, DistilBERT, LoRA, Llama Guard, BLIP*

- Evaluated and compared modeling approaches (LSTM baseline vs. LoRA-fine-tuned DistilBERT), tracking accuracy, precision, recall, and F1 across categories using Weights & Biases.
- Tested a zero-shot content moderation pipeline (Llama Guard) alongside a multimodal BLIP captioning module, weighing trade-offs between approaches ahead of Streamlit deployment.

TRAINING & CERTIFICATIONS

IBM-Certified AI & Data Science Professional | Digital Egypt Pioneers Initiative (DEPI) Nov 2024 – May 2025

- Completed an IBM-certified program covering Python, ML, SQL, and MLOps; built and evaluated ML/DL models, achieving 91.12% accuracy on the capstone project.

NLP Training | Neurova AI Apr 2026 – Jul 2026

- Completed an intensive 8-week NLP training program spanning classical preprocessing, TF-IDF, Transformer architectures, BERT fine-tuning, RAG, AI Agents and LLM integration across 7 end-to-end projects.